Original Articles: Phosphorus and Potassium Content of Enhanced Meat and Poultry Products: Implications for Patients Who Receive Dialysis

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Background and objectives: Uncooked meat and poultry products are commonly enhanced by food processors using phosphate salts. The addition of potassium and phosphorus to these foods has been recognized but not quantified. Design, setting, participants, & measurements: We measured the phosphorus, potassium, and protein content of 36 uncooked meat and poultry products: Phosphorus using the Association of Analytical Communities (AOAC) official method 984.27, potassium using AOAC official method 985.01, and protein using AOAC official method 990.03. Results: Products that reported the use of additives had an average phosphate-protein ratio 28% higher than additive free products; the content ranged up to almost 100% higher. Potassium content in foods with additives varied widely; additive free products all contained <387 mg/100 g, whereas five of the 25 products with additives contained at least 692 mg/100 g (maximum 930 mg/100 g). Most but not all foods with phosphate and potassium additives reported the additives (unquantified) on the labeling; eight of 25 enhanced products did not list the additives. The results cannot be applied to other products. The composition of the food additives used by food processors may change over time. Conclusions: Uncooked meat and poultry products that are enhanced may contain additives that increase phosphorus and potassium content by as much as almost two- and three-fold, respectively; this modification may not be discernible from inspection of the food label.